Quadratic-twist registry: metadata consistency

2026-10-22 — bounded correctness and metadata freshness

At PR head `3e442fd3310d`, the full native replay, validation gate, catalog assessment and this study's Lean target passed CI. The certificate job then rejected compact JSON rows at the metadata-only standards join. This follow-up fixes the Rust importer's appended-row layout and dependent metadata. Every registry value, model, subgroup, map identity and geometry finding is unchanged; arbitrary-size integer tokens remain exact.

Profile	Models	Same-field construction
Ordinary binary	165	genus 3, degree 3
Direct odd prime	2,011	genus 2, degree 2
Normalized odd prime	30	genus 2, degree 2; target normalization
Direct odd extension	2,287	genus 2, degree 2 over the declared extension
Total	4,493	4,491 distinct cover models

The graph still binds 538 declared fields, 4,491 Jacobian contexts, 4,493 maps, and 1,004 selected EC1 representations (140 binary, 864 odd-characteristic). The 942-model native twist study is separate. Its 434 deterministic comparisons, 84 IC matrix checks and 18 Lean theorems retain their original finite or explicitly conditional scope.

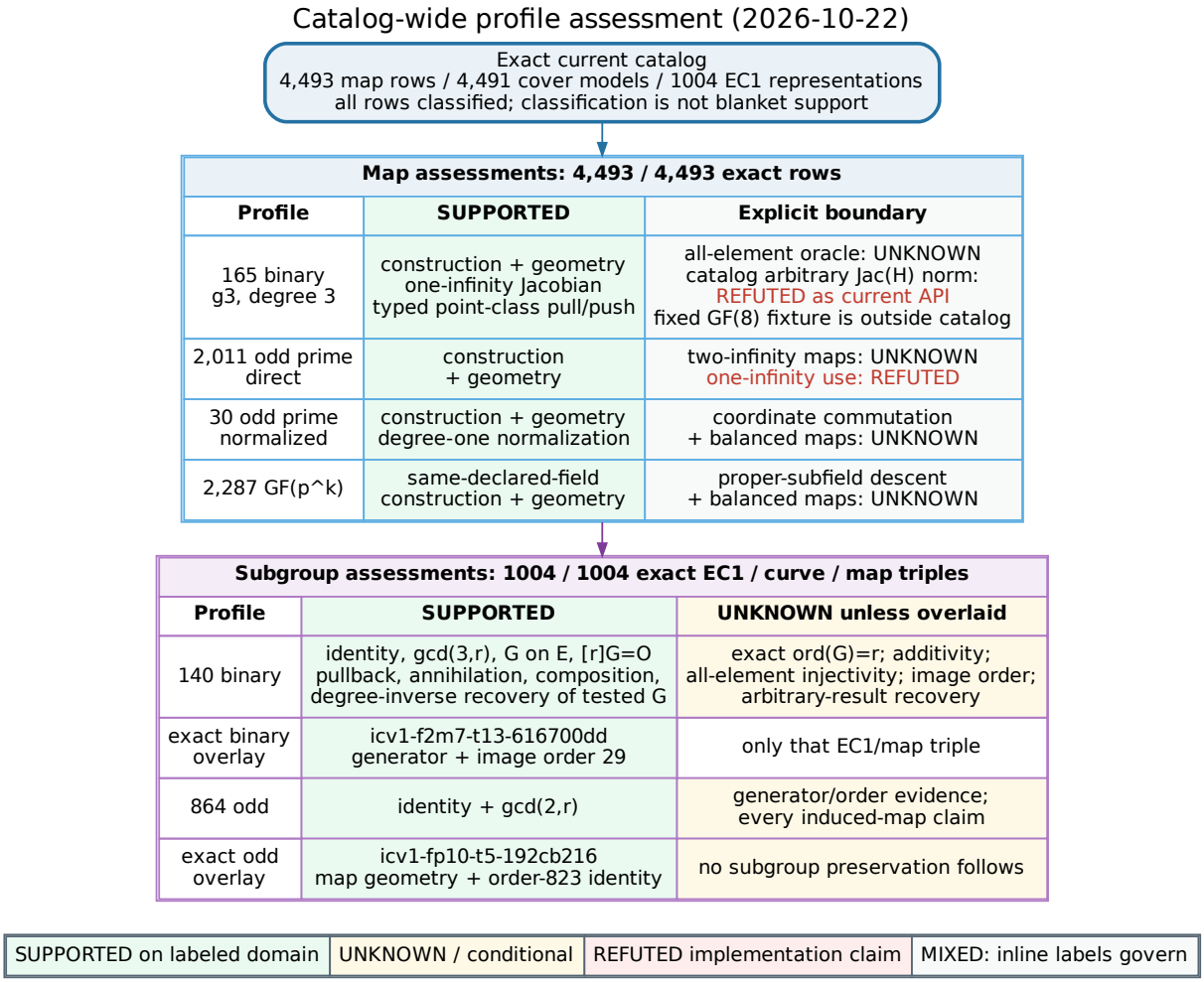
Evidence. All 40 native metadata/backend tests pass. Native identity certificates reproduce byte for byte; changed input is rejected before output. A fresh import from the original 2,572-model catalog matches the canonical 4,493-model registry byte for byte. The standards metadata join passes. Odd extension roster labels now display the actual characteristic and degree; all measured leaderboard values and protocols are preserved. The source-linked validation record contains the native cover, Cargo and assessment receipts. The browser now displays native integer coefficient vectors and all 4,493 models, preserving every prior curve row and measurement index.

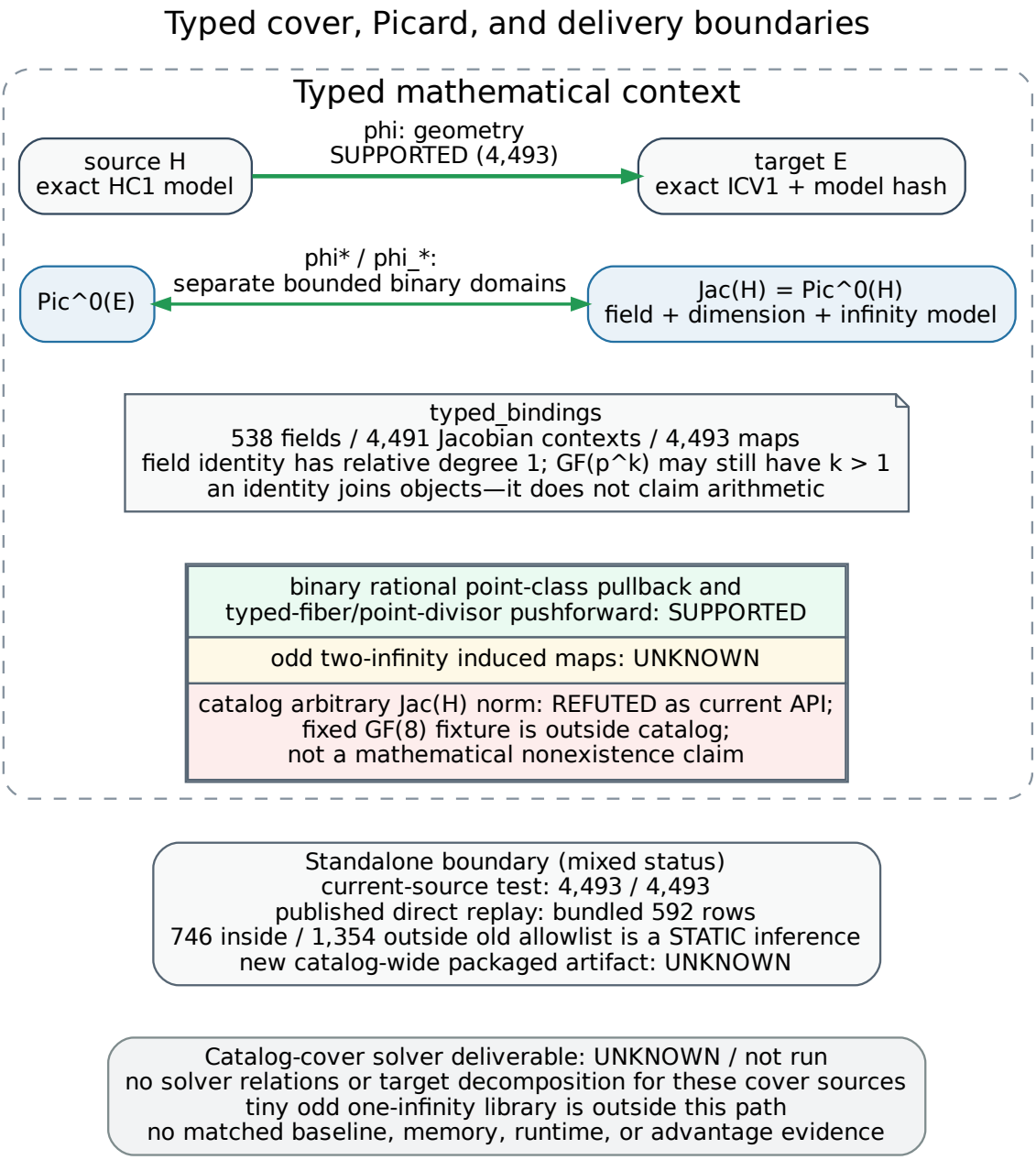
Provenance. The exact catalog blobs at `a11984a57344` remain required predecessor inputs. The active assessment pins a descendant snapshot with the layout correction and refreshed registry digests. Every existing formula, membership, partition and semantic check remains enabled. Historical study runs, earlier reports, failures and source archives remain unchanged.

Open gates. The unrelated SEA project builds, but Lean 4.19 rejects the global gate's `-trusted` argument before the axiom audit. Its sources are identical at the declared base, CI base, PR head and CI merge; this PR does not patch that target or gate. Cursor's approval agent could not start because its usage limit was reached. Current-head CI and review must still satisfy all applicable gates. Shared Conductor remains unavailable; no reservation is claimed.

The actual twist embedding's Frobenius-negation property and full eigenspace, summation-polynomial, pairing and trace-zero bridges remain open. No general hardness, GHS, speedup or invalid-curve-attack claim follows. The unchanged diagrams describe the same population and support boundaries. Canonical cost graphs are unchanged because no cost comparison was performed.

Map and subgroup boundaries




Typed cover, Picard, and delivery boundaries
Typed mathematical context
source H
exact HC1 model
phi: geometry
SUPPORTED (4,493)
target E
exact ICV1 + model hash
Pic^0(E)
phi* / phi_*:
separate bounded binary domains
Jac(H) = Pic^0(H)
field + dimension + infinity model
typed_bindings
538 fields / 4,491 Jacobian contexts / 4,493 maps
field identity has relative degree 1; GF(p^k) may still have k > 1
an identity joins objects—it does not claim arithmetic
binary rational point-class pullback and
typed-fiber/point-divisor pushforward: SUPPORTED
odd two-infinity induced maps: UNKNOWN
catalog arbitrary Jac(H) norm: REFUTED as current API;
fixed GF(8) fixture is outside catalog;
not a mathematical nonexistence claim
Standalone boundary (mixed status)
current-source test: 4,493 / 4,493
published direct replay: bundled 592 rows
746 inside / 1,354 outside old allowlist is a STATIC inference
new catalog-wide packaged artifact: UNKNOWN
Catalog-cover solver deliverable: UNKNOWN / not run
no solver relations or target decomposition for these cover sources
tiny odd one-infinity library is outside this path
no matched baseline, memory, runtime, or advantage evidence